Data import and display packages: Importing data and displaying them in appropriate ways is the most important task of data science. R has several popular packages related to the data import and display category.

- *readxl:* This is the most popular package to import data from Excel files. Designed and developed by Hadley Wickham, the main property of this package is to read Excel files in R speedily without any dependencies.
- *readr:* This package was also written by Hadley Wickham. It is best suited for huge files and to read CSV files faster. A similar package of this category is Vroom, developed by Jim Hester.
- ***rio**:* This was designed by Thomas J. Leeper. It supports the Web based import from SSL and HTTPS. Compressed files can also be read directly without explicit decompression.
- *datapasta:* Miles McBain is one of the authors of this package. If you have copied any data from the Web or a spreadsheet, and want to paste it into an R object so that it can be reproduced, this package will work for you.
- *httr:* This package is useful for pulling data from Web APIs. It provides functions for all important aspects of HTTP such as *GET(), HEAD(), PATCH(), PUT(), DELETE()* and *POST()*. It was designed by Hadley Wickham.

Data visualisation packages: In data science, these are the most crucial as they display the outcome in a pictorial form so that anybody can understand it. It is also very useful for exploratory data analysis. Some important data visualisation packages in R are listed below.

- *ggplot2:* This is the most powerful visualisation package in R. It is based on the grammar of graphics theory. With the help of this package, we can build custom plots more easily. It comprises two main functions — *qplot()* and *ggplot()*. It was designed by Hadley Wickham.
- *Lattice:* This package is good for multi-variate data. It is inspired by Trellis Graphics. Built using grid packages, it was written by Deepayan Sarkar.
- *highcharter:* This is an interactive visualisation package in R. It was designed by Joshua Kunst and is very useful for dynamic charting. This package has easy-to-customise themes for interactive visualisation.
- *Leaflet:* Joe Cheng, Bhaskar Karambelkar and Yihui Xie wrote this package. It is lightweight but powerful enough to build interactive maps.
- *RColorBrewer:* This is very handy to manipulate colours in plots, graphs and maps. Erich Neuwirth designed this package, using which you can design nice colour palettes.
- *plotly:* This is also an interactive visualisation package consisting of different categories of charts. The main attractions of this package are contour plots, candlestick charts and 3D charts.

We are not able to accommodate all the packages related to data science in this article but have included all the main ones essential to address the basic functionalities required in this field. END

References

[1] *https://www.guru99.com/data-science-tutorial.html*
[2] *https://www.trifacta.com/data-wrangling/*
[3] *https://cran.r-project.org/web/packages/available_packages_by_date.html*
[4] *https://mode.com/blog/r-data-visualization-packages*
[5] *https://www.computerworld.com/article/2921176/great-r-packages-for-data-import-wrangling-visualization.html*

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